symm

SYMBIOTIC MICROBES

THE SYMM PROGRAM

Field Trials & Research

The measured record behind the Symm program: yield, resilience under stress, and nutrient-use efficiency. Every figure here is labeled with the kind of trial or program behind it.

CHEMICAL-FREE | PESTICIDE-FREE | NON-GMO

REPLICATED FIELD TRIALS · PEER-REVIEWED RESEARCH ON THE ORGANISMS · AIDAN@SYMMBIOTIC.COM

Yield. Resilience. Efficiency.

Two parts, one program

Symm is one biological program with two parts, applied together as a seed, in-furrow or foliar treatment: one for resilience, one for nutrition. Part 1 and Part 2 are sold and applied together as one program; the components are not sold separately. The program has been developed over successive seasons, so the trials here span successive formulations of the two parts.

PART 1 · RESILIENCE

Resilience from within

A consortium of advanced *Trichoderma harzianum* strains that partner with the plant to raise tolerance to drought and heat, and to strengthen roots and establishment.

PART 2 · NUTRITION

Nutrient-use efficiency

A five-strain *Bacillus* consortium (*B. licheniformis, B. megaterium, B. pumilus, B. subtilis, B. amyloliquefaciens*) that unlocks soil-bound nutrients so the crop accesses more of the available nutrients.

The headline figures

+3.4%

Average yield gain over the untreated check, seed treatment with both parts, across 31 trial sets, 2020 to 2025 (median +2.5%). Every season from 2020 to 2025 averaged a gain.

27 of 31

Seed-treatment trial sets, 2020 to 2025, where treated out-yielded the untreated check (87%). The same 31 sets behind the average.

$p<0.0001$

Odds of 27 or more wins out of 31 by chance: under 1 in 10,000.

Headline figures: seed treatment with both parts, 31 trial sets, 2020 to 2025. How they are built is on the next page.

How to read the evidence

Every figure carries a label saying what kind of trial stands behind it.

LABEL	WHAT IT MEANS
REPLICATED FIELD	Replicated field trial run by a third-party research cooperator, with treated and untreated plots in replicate. The strongest evidence we hold.
FIELD TRIAL	Open-field trial run in-house or with a working farm, most as treated and untreated plots side by side rather than in replicate.
CONTAINER	Trial in pots, bins or trays, measuring establishment and growth under controlled conditions.

How the figures are built

> Each row-crop figure is one trial set: the average of its two-part seed-treatment arms, each compared with the untreated check at the same input level. The headline average, **+3.4%**, covers all 31 seed trial sets from 2020 to 2025. Each p-value compares every treated plot in a set with its untreated plots (Welch two-sample t-test); the odds on the headline page come from a sign test across all 31 sets.

Every figure in this document compares treated plots to untreated controls within the same trial, under the same management. Cooperator identities are held on file.

 REPLICATED FIELD

Corn, soybean and wheat

The core of the record. In replicated field seed trials from 2020 to 2023, Symm-treated plots out-yielded the untreated check in 19 of 23 corn, soybean and wheat trial sets.

CORN · 2020
+6.0%
p=0.04

SOYBEAN · 2021
+6.7%
p=0.11

WHEAT · 2021
+6.6%
p=0.003

WINTER WHEAT · 2020
+5.7%
p=0.02

EVERY CORN, SOYBEAN AND WHEAT SEED TRIAL SET, 2020 TO 2023

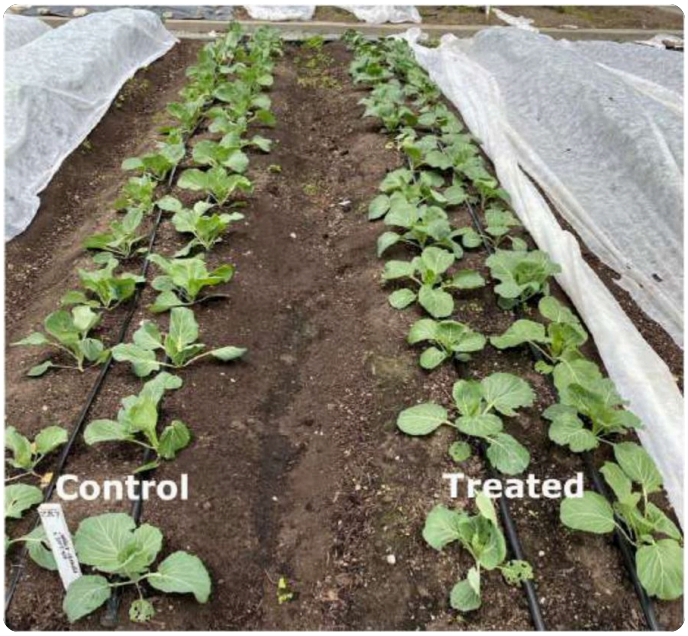
CROP	YEAR	P-VALUE	YIELD VS CONTROL
Corn	2020	0.04	**+6.0%**
Corn	2020	0.17	**+3.5%**
Corn	2020	0.44	**+0.7%**
Corn	2020	0.95	**−1.0%**
Corn	2021	0.38	**+5.6%**
Corn	2021	0.09	**+4.3%**
Corn	2021	0.53	**+2.5%**
Corn	2022	0.14	**+5.2%**
Corn	2023	0.66	**+1.9%**
Corn	2023	0.41	**+1.1%**
Corn	2023	0.56	**−1.5%**

CROP	YEAR	P-VALUE	YIELD VS CONTROL
Soybean	2020	0.30	**+5.4%**
Soybean	2021	0.11	**+6.7%**
Soybean	2021	0.14	**+2.5%**
Soybean	2022	0.89	**+1.2%**
Soybean	2023	0.87	**+1.7%**
Wheat	2020	0.82	**+1.1%**
Wheat	2021	0.003	**+6.6%**
Wheat	2021	0.45	**−3.9%**
Wheat	2022	0.12	**+4.8%**
Winter wheat	2020	0.02	**+5.7%**
Winter wheat	2020	0.12	**+2.7%**
Winter wheat	2020	0.96	**−0.2%**

Third-party research cooperators, 2020 to 2023. Cooperator identities are held on file. Taken together, 27 of 31 seed trial sets out-yielded the check, odds under 1 in 10,000 by chance.

Specialty and vegetable crops

Treated plots gained in germination, vigor, size and quality across a wide range of vegetable and specialty crops. In all 14 specialty observations, across 11 crops and a mixed-species seedling study, the treated side came out ahead.



Control and treated rows, marked in the same bed.

Cabbage · harvest weight

+24.8%

Control and treated rows in the **same bed**, same planting date, same irrigation, so the comparison is visible in the photograph.

FIELD TRIAL



Lettuce **+14.9%**

Rows under drip.

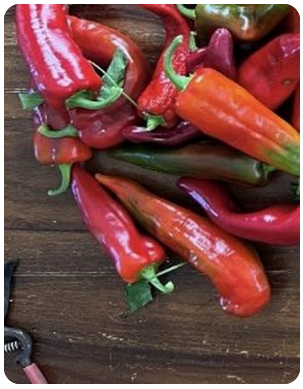
FIELD TRIAL



Snap peas **+34.9%**

Full canopy.

FIELD TRIAL



Peppers **+91%**

Fruit weight, single trial.

FIELD TRIAL



Peppers **+64%**

Fruit count, same trial.

FIELD TRIAL

Open-field trials run in-house and with working farms, 2020 to 2025, mostly as side-by-side plots. Percentages compare treated to untreated controls within each trial.

Stronger starts

Treated seed germinated faster and more completely across the seedling studies, including after drought.

Spinach · germination trial, relative gain **+28%**



CONTROL 36 OF 50



TREATED 46 OF 50

Same tray design, same sowing date, grown side by side. In a separate drought trial, viability rose from 85.7% to 95.6%. GERMINATION

GERMINATION & ESTABLISHMENT

- Sugar snap peas germinated **46% faster** than untreated seed.
- Spinach viability rose **85.7% to 95.6%** after drought; a seed trial showed **+28% relative germination**.
- Onion seed held one year at 35°C germinated at **50% against 0%** untreated.
- **+53.7% seedling biomass** (mixed species) after one week of drought.

GERMINATION Seedling-tray studies of establishment under stress.

Seedling-tray studies, 2020 to 2025. Each compares treated with untreated seed sown side by side.

The full specialty record

Across fruiting vegetables, brassicas, alliums and cucurbits, treated plots out-yielded untreated controls, often by wide margins over a full season of harvest. Season-harvest figures are cumulative across the harvest window, which is why they run higher than a single-date yield comparison.

CROP	MEASURE	VS CONTROL	TRIAL TYPE
Cucumber	Season harvest, p<0.0062	**+148.8%**	FIELD TRIAL
Kale	Season harvest weight	**+111%**	FIELD TRIAL
Peppers	Fruit weight / count, Cubanelle, single trial	**+91% / +64%**	FIELD TRIAL
Winter squash	Harvest, six varieties	**+12.8 to +46.8%**	FIELD TRIAL
Snap peas	Harvest weight	**+34.9%**	FIELD TRIAL
Garlic	Bulb weight / bulb size	**+7.6% / +8.2%**	FIELD TRIAL
Cabbage	Harvest weight	**+24.8%**	FIELD TRIAL
Lettuce	Harvest weight	**+14.9%**	FIELD TRIAL
Tomato	Harvest weight / per fruit	**+8.4% / +5.4%**	CONTAINER

Open-field and container trials, 2020 to 2025. Each row is a single trial; percentages compare treated to untreated controls within that trial, and results vary with crop, season and site.

07 THE MECHANISM

Why it works

Two organisms doing two different jobs, both of them well documented in peer-reviewed research that predates Symm. This page is the science rather than trial data.

PART 2 · BACILLUS

Unlocks soil nutrients

The *Bacillus* consortium solubilizes soil-bound phosphorus and micronutrients through organic-acid pathways, making them available to the plant.

PART 2 · UPTAKE

Improves efficiency

Better nutrient-use efficiency helps the crop access more of the available nutrients. It does not replace a fertility program, which should continue as normal.

PART 1 · TRICHODERMA

Deeper, stronger roots

The *Trichoderma* partnership drives root development, widening the volume of soil each plant can feed from and reach water in.

What the research on these organisms shows

MEASURE	DIRECTION WITH TREATMENT
Nutrient uptake and nutrient-use efficiency	**Higher**
Root colonization and root development	**Higher**
Seed and seedling tolerance to drought and heat	**Higher**
Phosphorus and micronutrients freed from the soil	**Higher**

Each row is established in the peer-reviewed research listed on page 10.

On fertilizer. Symm improves how efficiently a crop uses the nutrients available to it. It is not a fertilizer, not a replacement for one, and we make no claim that it allows a fertility program to be reduced. Growers on Symm should maintain their normal fertility program.

Peer-reviewed sources are listed on page 10.

More crops, more yield

The same program carries well beyond grain and vegetables, into forage, fiber and specialty crops, applied as a seed treatment or a foliar spray. Counting all wheat types as one crop, every one of the 11 crops tested with both parts averaged a gain over the untreated check, across 50 seed, foliar and in-furrow comparisons, 43 of which beat the check. Corn, soybean and wheat are on page 4; the other eight are below.

CROP	MEASURE	VS CONTROL	TRIAL TYPE
Hemp	Yield, foliar / seed treatment	**+18.6% / +14.7%**	FIELD TRIAL
Rice	Yield, seed treatment, four trials, all gains	**+0.4 to +29.7%**	REPLICATED FIELD
Alfalfa	Forage yield, five comparisons, range −18.0 to +18.0%	**+4.0% average**	REPLICATED FIELD
Sorghum	Yield, seed treatment, one trial	**+2.5%**	REPLICATED FIELD
Potato	Yield, foliar, one trial set	**+5.6%**	REPLICATED FIELD
Turf	Yield, foliar and in-furrow, two trial sets	**+4.2 to +7.9%**	REPLICATED FIELD
Field peas	Yield, seed treatment, one trial set	**+10.7%**	FIELD TRIAL
Oats	Yield, seed treatment, one trial set	**+12.1%**	FIELD TRIAL

SEED OR FOLIAR, BOTH WORK

Symm delivered gains whether applied as a seed treatment or a foliar spray, which gives growers flexibility to fit the program to their operation and their equipment rather than the other way around.

TESTED ACROSS SEASONS

Seed treatment averaged a gain in every season from 2020 to 2025, so the results reflect performance through varied weather and soils rather than a single favorable season.

> Rows show the two-part program as trialed, without added products.

Results from replicated and open-field trials; figures compare Symm-treated to untreated controls within each trial. Trial type is labeled on every row.

The peer-reviewed research

Symm's two parts rest on decades of peer-reviewed research into the two classes of microbes at their core, from Cornell University and other leading institutions.

Trichoderma harzianum · the resilience science

The research below concerns the genus *Trichoderma*, not Symm's specific strains. What it establishes is the mechanism: that *Trichoderma* colonizes roots, improves nutrient acquisition, and raises stress tolerance, in controlled and peer-reviewed work. Symm Part 1 is a consortium selected within that genus. The literature establishes the mechanism; the trials in the preceding pages show the result in the field.

REFERENCE	WHAT IT ESTABLISHES
Harman, Howell, Viterbo, Chet & Lorito Nature Reviews Microbiology, 2004	The landmark review establishing *T. harzianum* as a partner to the plant that improves root colonization, stress tolerance and nutrient-use efficiency across both monocots and dicots.
Mastouri, Björkman & Harman Phytopathology, vol. 100, 2010	A low-cost seed treatment confers measurable tolerance to drought and heat in germinating seed and seedlings.
Mastouri, Björkman & Harman Molecular Plant-Microbe Interactions, 2012	Molecular evidence that *T. harzianum* maintains plant redox balance under water deficit, at or above the level of unstressed plants.
Harman Phytoparasitica, vol. 39, 2011	Yield gains and improved nutrient-use efficiency across crops.

Bacillus · the nutrition science

AREA	WHAT THE LITERATURE ESTABLISHES
Phosphate and micronutrient solubilization	Peer-reviewed work on how *Bacillus* species free soil-bound phosphorus and micronutrients for plant uptake.
Plant-growth promotion	Reviews of *B. subtilis* and *B. amyloliquefaciens* as growth promoters and biofertilizer organisms.
Stress resilience	*Bacillus* as plant probiotics supporting growth and resilience in adverse conditions.

Full citations available on request.

symm

SYMBIOTIC MICROBES

THE TAKEAWAY

Resilience you can measure

One two-part biological program, sold and applied together. As a seed treatment it out-yielded the untreated check in **27 of 31** trial sets, an average gain of **+3.4%**, and every season from 2020 to 2025 averaged a gain. Every one of the 11 crops tested with both parts averaged a gain (wheat types counted together), the treated side came out ahead in all 14 specialty observations, and peer-reviewed research stands behind the organisms.

+3.4%

Average seed-treatment yield gain over untreated, 31 trial sets, 2020 to 2025 (median +2.5%).

27 of 31

Of those trial sets, treated out-yielded untreated (87%).

6 of 6

Seasons from 2020 to 2025 in which seed treatment averaged a gain.

> **Where the record stands.** The trials in this document cover row, forage, fiber and vegetable crops.

CHEMICAL-FREE · PESTICIDE-FREE · NON-GMO

Yield. Resilience. Efficiency.

Symm is a single two-part program, sold and applied together. Results are from replicated field trials, open-field and container trials, and peer-reviewed research on the organisms, labeled throughout. Individual farm results vary with soil, season and management. Nothing here is a guarantee for a given field.

Source Technologies LLC · 149 Franklin St, Floor 4, New York, NY 10013 · symmbiotic.com · aidan@symmbiotic.com · 845 797 3424

Symm Field Trials and Research · version of 27 October 2026 · Figures are updated as trials complete. Please check you have the current version before relying on it.